



Time Series

Basic Statistical Concepts

Basic Statistical Concepts

- Moments of a Distribution: Fully characterizing the distribution
- Estimation Methods: Method of Moments versus Maximum Likelihood Estimation
- Basic Estimators: Approach and Sampling Distribution
- Multivariate data: Joint, marginal, and conditional distribution
- Statistical Inference: Confidence intervals and Hypothesis Testing

Moments of a Distribution

- Moments of a random variable X with density $f(x)$:
- l -th moment: $m'_l = E[X^l] = \int_{-\infty}^{\infty} x^l f(x) dx$
- l -th central moment: $m_l = E[(X - \mu)^l] = \int_{-\infty}^{\infty} (x - \mu)^l f(x) dx$
- Examples of moments:
- Expectation $E[X]$
- Variance $E[(X - \mu)^2]$
- Skewness $S(x) = E\left[\frac{(X - \mu)^3}{\sigma^3}\right]$
- Kurtosis $K(x) = E\left[\frac{(X - \mu)^4}{\sigma^4}\right]$

Statistical Estimation

Parametric Statistics: Observe $x_1, \dots, x_n$ (realizations) from a set of random variables $X_1, \dots, X_n \sim f(x; \theta)$ where $f(x; \theta)$ is a density function with parameter θ which is assumed unknown

Estimation: evaluate the unknown parameter θ using the set of observations $x_1, \dots, x_n$ and using the distribution of the random variables $X_1, \dots, X_n$ from which we observe.

Approaches:

- 1. Method of Moments (MOM)
- 2. Maximum Likelihood Estimation (MLE)

Examples of Classic Estimators

- Given the data $\{X_1, \dots, X_n\}$, estimate mean, variance, skewness and kurtosis with
- *Sample mean:* $\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
- *Sample Variance:* $\hat{\sigma}^2 = S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$
- *Sample skewness:* $\hat{S}(X_1, \dots, X_n) = \frac{1}{(n-1)\hat{\sigma}^3} \sum_{i=1}^n (X_i - \hat{\mu})^3$
- *Sample kurtosis:* $\hat{K}(X_1, \dots, X_n) = \frac{1}{(n-1)\hat{\sigma}^4} \sum_{i=1}^n (X_i - \hat{\mu})^4$

Sampling Distributions of Estimators

- Sampling Distributions: Under normality assumption
- $X_1, \dots, X_n \sim N(\mu, \sigma^2)$
- $\bar{X} \sim N(\mu, \sigma^2/n)$
- $\frac{s^2(n-1)}{\sigma^2} \sim \chi_{n-1}^2$
- Properties of the Estimators
- 1. Unbiasedness
- 2. Consistency

Method of Moments Estimation

- **MOM Approach:** Equate the distribution moments to the observed moments
- $E[X^p] = \frac{1}{n} \sum_{i=1}^n X_i^p$
- Example: $X_1, \dots, X_n \sim N(\mu, \sigma^2)$
- $(p = 1) : E[X](= \mu) = \frac{1}{n} \sum_{i=1}^n X_i \Rightarrow \hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$
- $(p = 2) : E[X^2](= \sigma^2 + \mu^2) = \frac{1}{n} \sum_{i=1}^n X_i^2 \Rightarrow \hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$

Maximum Likelihood Estimation

- **MLE Approach:** Maximize the likelihood function of θ given the data.
- Joint distribution of $X_1, \dots, X_n \sim f(x; \theta)$ under independence assumption
- $f(x_1, \dots, x_n; \theta) = f(x_1; \theta) \dots f(x_n; \theta)$
- Likelihood function
- $L(\theta : x_1, \dots, x_n) = f(x_1, \dots, x_n; \theta)$

$$\text{MLE: } \hat{\theta} = \operatorname{argmax}_{\theta \in \Theta} L(\theta : x_1, \dots, x_n)$$

Multivariate Distribution

- **Joint, Marginal and Conditional distributions:**

- *Joint distribution = Conditional $\times$ Marginal:* $f(x, y) = f(x|y)f(y) = f(y|x)f(x)$

- For three variables X, Y, Z :

- $f(x, y, z) = f(x|y, z)f(y, z) = f(x|y, z)f(y|z)f(z)$

- For n variables $X_1, \dots, X_n$:

- $f(x_1, \dots, x_n) = f(x_n|x_{n-1}, \dots, x_1)f(x_{n-1}|x_{n-2}, \dots, x_1) \dots f(x_2|x_1)f(x_1)$

- Example: if $X_i|X_{i-1}, \dots, X_1$ is normal with mean μ_i and variance σ_i^2 and if $f(x_1)$ is ignored

- $$f(x_n, x_{n-1}, \dots, x_1) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_i}} \exp \left[\frac{-(x_i - \mu_i)^2}{2\sigma_i^2} \right]$$

Statistical Inference

- **Hypothesis Testing**
- Parameter-based hypothesis testing: $H_0: \theta = \theta_0$ vs $H_a: \theta \neq \theta_0$
- Distribution-based hypothesis testing:
 - $H_0: X_1, \dots, X_n \sim N(\mu, \sigma^2)$ vs $H_a: \text{non-normal distribution}$
 - *P-value* = a measure of the plausibility of H_0
 - *Significance Level* = the probability of type 1 error (reject H_0 when it is true)
- **Confidence Interval** A $(1 - \alpha)$ confidence interval for θ is $(\hat{\theta} - k, \hat{\theta} + k)$ with k s.t.
- $\Pr(\theta \in (\hat{\theta} - k, \hat{\theta} + k)) = 1 - \alpha$